Ömer Gölcük 38101 **EEE 58003 - VLSI System Design I**

Lab3 Report

Introduction

In this laboratory, a pipelined multiply–accumulate (MAC) unit was designed, implemented, and analyzed using a full RTL-to-gate-level digital design flow. The MAC unit consists of a 16×16 Wallace tree multiplier followed by a 36-bit accumulator, where the final addition is performed using a hybrid Kogge–Stone and ripple-carry adder structure. The design was first verified functionally using RTL simulation to ensure correctness of the arithmetic and pipeline behavior. Subsequently, the design was synthesized using Synopsys Design Vision with a TSMC 65 nm standard-cell library under a 250 MHz timing constraint. Post-synthesis timing analysis was performed to evaluate critical paths and slack, and gate-level simulation with SDF annotation was conducted in SimVision to validate correct operation under realistic propagation delays. This workflow demonstrates both functional correctness and timing-aware verification of a realistic hardware data-path.

Design of The Multiplier Accumulator (MAC) Unit

The MAC unit was designed as a pipelined data-path consisting of a multiplier stage followed by an accumulation stage. The multiplication is performed using a 16×16 Wallace tree multiplier, where partial products are generated in parallel and reduced efficiently using carry-save adders to minimize the critical path. The final two rows of the Wallace tree are summed using a fast parallel-prefix adder. For this purpose, a 32-bit Kogge–Stone adder was employed for the lower bits to achieve high performance, while the upper 4 bits were handled by a ripple-carry adder to reduce area and design complexity. The accumulation stage uses a 36-bit adder, where the 32-bit product is zero-extended and added to the accumulator register each cycle. Pipeline registers are placed at the multiplier input and output, allowing the design to achieve high operating frequency while maintaining correct sequential behavior. The following schematic diagram shows the constructed MAC architecture.

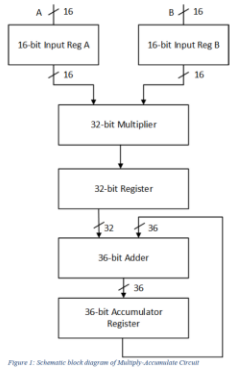

Figure 1: Architecture of MAC unit

The operation principle of the MAC unit is simple, in one clock cycle the Multiplicand A (16-bit) and Multiplier B (16-bit) is fed to the Wallace Tree multiplier and result is stored in output register to be used by 36-bit adder in the next clock cycle. Therefore, after two clock cycles the result will be accumulated in "Acc_out" register that is defined in my RTL design. The sequential (pre-synthesis behavior of the circuit can be seen in the following figure.

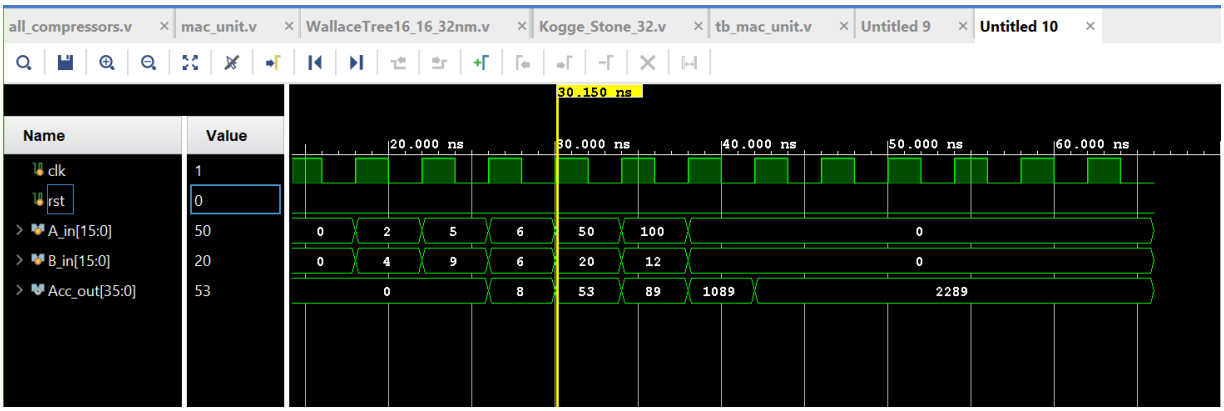

Figure 2:Pre-synthesis behavior of the MAC unit

If one looks closely, in can be seen that the accumulator register follows the result of the multiplication in 2 clock cycle delay.

Synthesis of the MAC unit in Synopsys Design Vision

After functional verification at the RTL level, the MAC unit was synthesized using Synopsys Design Vision with a TSMC 65 nm standard-cell library. A target clock frequency of 250 MHz (4 ns period) was applied as the primary timing constraint. Post-synthesis static timing analysis was performed to evaluate the critical paths of the design and verify timing closure. The results indicate positive slack under the specified clock constraint, confirming that all setup timing requirements are met. Based on the reported slack and clock period, the critical path delay corresponds to a maximum achievable operating frequency significantly higher than the target frequency, demonstrating that the pipelined MAC architecture is capable of reliable operation at 250 MHz.

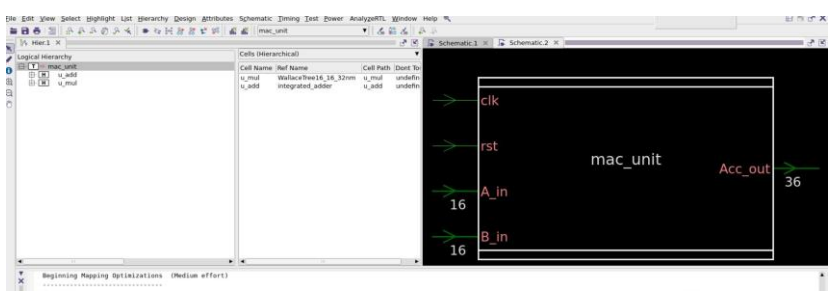

Figure 3:The Schematic of the MAC Unit in Design Vision

Post-synthesis static timing analysis was performed using Synopsys Design Vision under a 250 MHz clock constraint (4.00 ns period). The critical path originates from the product register (P_reg_reg[26]) and propagates through the combinational logic to the destination register clock pin. The data arrival time on this path was measured as 2.18 ns, while the data required time, considering the clock period and library setup time, was 3.97 ns. This results in a positive setup slack of **1.79 ns**, indicating that the design comfortably meets the specified timing constraint. The presence of substantial positive slack confirms that the MAC unit is not timing-critical at 250 MHz and can potentially operate at higher clock frequencies. The following figure shows related result obtained @ 250 MHz.

```
  U240/ZN (INR2D1)                                                    0.06       2.18 r
  P_reg_reg[26]/D (DFQD1)                                             0.00       2.18 r
  data arrival time                                                              2.18

  clock clk (rise edge)                                               4.00       4.00
  clock network delay (ideal)                                         0.00       4.00
  P_reg_reg[26]/CP (DFQD1)                                            0.00       4.00 r
  library setup time                                                 -0.03       3.97
  data required time                                                             3.97
  -------------------------------------------------------------------------------------
  data required time                                                             3.97
  data arrival time                                                             -2.18
  -------------------------------------------------------------------------------------
  slack (MET)                                                                    1.79


1
design_vision>
```

Figure 4:Slack of 1.79 is obtained in for designed MAC unit

Design works like a charm up to 450 MHz, then got violated as one can see in the following figure.

```
  data arrival time                                                              2.18

  clock clk (rise edge)                                               2.20       2.20
  clock network delay (ideal)                                         0.00       2.20
  P_reg_reg[26]/CP (DFQD1)                                            0.00       2.20 r
  library setup time                                                 -0.03       2.17
  data required time                                                             2.17
  -------------------------------------------------------------------------------------
  data required time                                                             2.17
  data arrival time                                                             -2.18
  -------------------------------------------------------------------------------------
  slack (VIOLATED)                                                              -0.01


1
design_vision>
```

Figure 5:Slack of -0.01 is obtained for 450 MHz

After synthesis using the TSMC 65 nm standard-cell library (tcbn65lptc_ccs), the MAC unit consists of 3,412 standard cells, including 2,519 combinational cells and 100 sequential cells, with no macro or black-box components. The synthesized design contains 6,757 ports and 9,324 nets, indicating a moderately complex data-path dominated by arithmetic logic. The total combinational area is 7,146.36 µm², while the non-combinational (sequential) area is 684.00 µm², resulting in a total cell area of approximately 7,830.36 µm². Buffer and inverter cells contribute 34.56 µm² to the overall area. No net interconnect area is reported due to the zero wire-load model assumption. These results demonstrate that the MAC unit achieves an efficient area footprint while meeting the required performance constraints at 250 MHz. These results can be seen in the following figure,

```
Library(s) Used:

    tcbn65lptc_ccs (File: /storage/Technologies/TSMC/TSMC65NM/TSMC65nm/TSMCHOME/digital/Front_End/timing_power_noise/CCS/tcbn65lp_200a/tcbn65lptc_ccs.db)

Number of ports:                          6757
Number of nets:                           9324
Number of cells:                          3412
Number of combinational cells:            2519
Number of sequential cells:                100
Number of macros/black boxes:                0
Number of buf/inv:                          32
Number of references:                        5

Combinational area:                7146.360000
Buf/Inv area:                        34.560001
Noncombinational area:              684.000015
Macro/Black Box area:                 0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                   7830.360015
Total area:                      undefined
design_vision> uplevel #0 { report_power -analysis_effort low }
Loading db file '/storage/Technologies/TSMC/TSMC65NM/TSMC65nm/TSMCHOME/digital/Front_End/timing_power_noise/CCS/tcbn65lp_200a/tcbn65lptc_ccs.db'
```

Figure 6:Area analysis of the design mac unit in design vision

Power analysis of the synthesized MAC unit was performed using Design Vision with the TSMC 65 nm (tcbn65lptc_ccs) standard-cell library under a 250 MHz clock frequency. The total estimated power consumption is 0.653 mW, composed of 0.5213 mW internal power, 0.1313 mW switching power, and 427.6 nW leakage power. As expected, dynamic power (internal + switching) dominates the total consumption, and it scales proportionally with the operating frequency; therefore, the selected 250 MHz clock directly contributes to the observed switching and clock network power. The combinational logic is the primary power consumer, accounting for 0.3809 mW (58.34%), mainly due to the Wallace tree multiplier and the Kogge-Stone adder. The clock network contributes 0.2358 mW (36.11%), highlighting the overhead of high-frequency clock distribution, while registers account for 5.55% of the total power.

Unlike pre-synthesis (RTL-level) power estimates, which rely on idealized models and rough activity assumptions, this analysis is post-synthesis, meaning it is based on an actual gate-level netlist mapped to a real standard-cell library with characterized delays, capacitances, and power models. Consequently, the reported power values more accurately reflect realistic implementation behavior, including clock buffering, gate sizing, and leakage effects. These results demonstrate that the MAC unit achieves a balanced trade-off between performance and power efficiency while meeting the targeted timing constraints. These facts can be seen in the following figure:

```
                 Internal         Switching           Leakage            Total
Power Group      Power            Power               Power              Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------------
io_pad             0.0000           0.0000            0.0000            0.0000  (   0.00%)
memory             0.0000           0.0000            0.0000            0.0000  (   0.00%)
black_box          0.0000           0.0000            0.0000            0.0000  (   0.00%)
clock_network      0.2358           0.0000            0.0000            0.2358  (  36.11%)  i
register       2.8756e-02       7.4838e-03           31.9331        3.6272e-02  (   5.55%)
sequential         0.0000           0.0000            0.0000            0.0000  (   0.00%)
combinational      0.2567           0.1238          395.6326            0.3809  (  58.34%)
--------------------------------------------------------------------------------------------------
Total              0.5213 mW        0.1313 mW       427.5658 nW         0.6530 mW
design_vision>
```

Figure 7:Power analysis of the mac unit in design vision

Post-Synthesis with SimVision

After logic synthesis, the MAC unit was verified using post-synthesis gate-level simulation in Cadence SimVision with Standard Delay Format (SDF) back-annotation generated by Design Vision. Unlike RTL simulation, this stage operates on the synthesized gate-level netlist, where all combinational and sequential elements are mapped to real standard cells from the TSMC 65 nm library and include realistic propagation delays. The simulation was executed at the target 250 MHz clock frequency (4 ns period), ensuring that functional correctness is maintained under timing-accurate conditions. This step validates that the design not only behaves correctly at the RTL level but also remains functionally correct once physical delays and clocking effects are introduced.

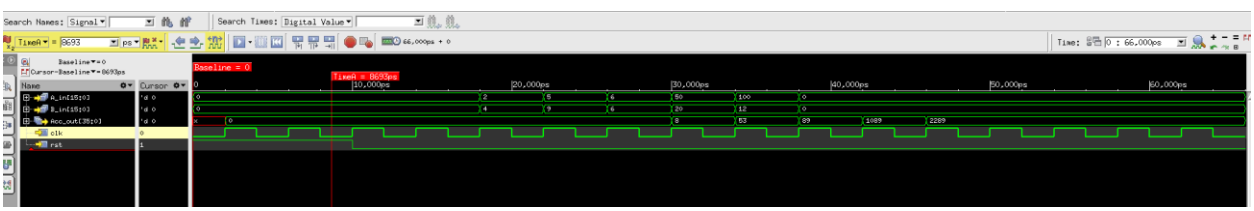

Figure 8: Waveforms Analysis of Post-Synthesis Simulation

The SimVision waveform illustrates the correct operation of the MAC unit under post-synthesis conditions. The input operands A_in and B_in are applied sequentially, while the accumulator output Acc_out updates after the expected pipeline latency introduced by internal registers and arithmetic stages. The waveform shows that multiplication results are correctly accumulated over successive clock cycles, and the final accumulated value converges to the expected total after the pipeline is flushed. Despite the presence of gate-level delays, no functional mismatches or timing violations are observed, confirming that the synthesized design satisfies both functional correctness and timing requirements at the specified operating frequency.

Conclusion

In this work, a fully pipelined MAC unit was designed, synthesized, and verified using both RTL and post-synthesis simulation flows. At the RTL level, functional simulation in Vivado confirmed correct arithmetic behavior and proper accumulation across clock cycles under ideal, zero-delay assumptions. Following synthesis with Design Vision using the TSMC 65 nm standard-cell library, the design was further validated through gate-level post-synthesis simulation in SimVision with SDF back-annotation, incorporating realistic cell and interconnect delays. The post-synthesis waveforms closely matched the RTL behavior, with differences limited to expected timing shifts due to propagation delays, demonstrating consistency between functional and timing-accurate models. Timing analysis showed a positive slack of 1.79 ns at 250 MHz, confirming that the design comfortably meets the target frequency. Overall, the MAC unit achieves correct functionality, timing closure, and efficient implementation, validating the chosen Wallace-tree multiplier and hybrid Kogge–Stone/RCA adder architecture under realistic operating conditions.